

Abstract

This project presents a real-time simulation and control system for a water tank using fuzzy logic implemented in LabVIEW. The primary objective is to regulate the water level by dynamically adjusting the inlet flow rate to overcome non-linear dynamics and system uncertainties. The controller accepts two input variables: the rate of change of water height (dH/dT) and the tank level error. Based on linguistically defined IF-THEN rules, the inference engine computes the required flow adjustment. The system was configured using LabVIEW's Fuzzy System Designer with triangular membership functions (five terms per input, seven for the output) and Center of Area (Centroid) defuzzification. Simulation results demonstrate that the controller successfully stabilizes the tank level at the 2.5-unit setpoint with a smooth transient response and zero steady-state error. These findings validate fuzzy logic as a robust, model-free alternative for real-time industrial process control.

Keywords: Real-Time Systems, LabVIEW, Water Tank Control, Fuzzy Logic, Membership Functions, Defuzzification

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1. Introduction



Figure 1: Visit to National Water Company

A field visit to the National Water Company revealed practical challenges in maintaining precise water levels in industrial storage tanks, where operators frequently rely on linguistic assessments rather than exact numerical thresholds. This observation motivated the selection of this project to explore fuzzy logic control as an intelligent alternative to conventional PID controllers, which often struggle with the non-linearities and uncertainties inherent in fluid systems (Ogata, 2010). Fuzzy logic enables control decisions based on linguistic rules that mimic human reasoning, making it particularly suitable for systems difficult to model mathematically (Ross, 2010). This project implements a real-time fuzzy logic controller using LabVIEW (National Instruments, 2023) to investigate whether such a controller can effectively regulate water tank levels without requiring precise mathematical modeling. The central hypothesis is that linguistic IF-THEN rules encoding expert knowledge will handle system uncertainties more effectively than traditional model-based approaches. Key challenges addressed include designing appropriate membership functions for non-linear tank dynamics, developing a comprehensive rule base covering all operating conditions, and ensuring real-time performance with minimal computational overhead. The expected gains from this work include a robust, intuitive control strategy applicable to industrial automation and practical competency in LabVIEW-based fuzzy system design—skills directly transferable to professional engineering contexts in process control and real-time systems.

2. Explanation

Fuzzy logic, introduced by Lotfi A. Zadeh in 1965, extends classical Boolean logic by allowing truth values to range continuously between 0 and 1. The term "fuzzy" refers to the inherent vagueness and imprecision of real-world phenomena. Rather than forcing every input into a crisp true/false classification, fuzzy logic accommodates partial membership, enabling more nuanced and robust decision-making under uncertainty (Zadeh, 1965). A fuzzy logic control system is built around four core components:

- **Rule Base:** Contains the set of IF-THEN rules defined by domain experts using linguistic terms. Each rule maps a combination of input states to an output action.
- **Fuzzification:** Converts crisp numerical inputs (e.g., sensor readings such as temperature, pressure, or flow rate) into fuzzy sets by evaluating their degree of membership in each linguistic category.
- **Inference Engine:** Evaluates the matching degree of current inputs against each rule in the rule base, fires the applicable rules, and combines their outputs to form an aggregate fuzzy control action.
- **Defuzzification:** Converts the aggregate fuzzy output set back into a single crisp value that can be sent to an actuator. Common methods include Center of Area (Centroid), Mean of Maxima, and Bisector; in this project the Center of Area method was employed (Ross, 2010).

Previous research on industrial tank level control has extensively compared classical PID architectures with intelligent control systems. Studies by Passino and Yurkovich (1998) and Ogata (2010) demonstrated that while PID controllers require precise mathematical modeling and extensive gain tuning, fuzzy logic controllers excel in non-linear, uncertain environments by leveraging qualitative operator knowledge. These studies consistently show that fuzzy systems achieve comparable or superior stability without relying on exact differential equations, directly validating our hypothesis that linguistic rules can effectively replace rigid mathematical models in fluid control applications.

To address our problem, we adopt the analysis techniques validated in this literature: overlapping triangular membership functions for computational efficiency, Min-Max inference for conservative rule firing, and centroid defuzzification for smooth, oscillation-free control output. In LabVIEW, the Fuzzy System Designer tool provides a graphical interface to define input and output membership functions and to construct the rule base. The resulting fuzzy system is saved as a .fs file and called from the LabVIEW block diagram via the FL Fuzzy Controller VI, making integration into a real-time simulation straightforward. This architecture directly supports our workflow: the real-time rule weights array enables transparent visualization of the controller's decision-making process, while the process monitor facilitates continuous data analysis and performance evaluation against our setpoint tracking hypothesis. By following established fuzzy control methodologies, we ensure that our data acquisition, visualization, and evaluation phases are grounded in proven engineering practices.

3. Problem Statement / Hypotheses

Briefly The problem addressed in this project is the automatic regulation of water level in a single-tank system. The tank receives an inlet flow controlled by a valve and drains through an orifice at its base. The objective is to design a fuzzy logic controller that drives the water level to a desired setpoint and maintains it despite the non-linear outflow dynamics governed by Torricelli's law (Ogata, 2010).

The physical model of the tank is described by the following governing equations:

1. **Outflow:** $f_{out} = A_o / 10 \times (2 \times 9.81 \times h)^{1/2}$
2. **Rate of height change:** $dh/dt = (f_{in} - f_{out}) / A_t$
3. **New height:** $h_{new} = h + dh/dt \times dt$

where A_o is the orifice area, A_t is the tank cross-sectional area, f_{in} is the inlet flow rate, and dt is the simulation time step.

The dataset and key parameters were configured based on standard laboratory specifications and practical insights gathered during a field visit to the National Water Company. Important parameters include: tank area ($A_t = 100$ units²), orifice area ($A_o = 9$ units²), simulation time step ($dt \approx 0.019$ s), input ranges for tank error and rate of change (-0.01 to 0.01), output range for flow change (-0.5 to 0.5), and target setpoint (2.50 units). The controller inputs are the tank level error (setpoint – actual level) and the rate of change of height (dH/dT); the output is the flow change applied to the inlet valve. Membership functions and rule definitions were configured using LabVIEW's Fuzzy System Designer (National Instruments, 2023; <https://www.ni.com/docs/en-US/bundle/labview-fuzzy-logic-toolkit/>).

The hypothesis is that a properly tuned fuzzy logic controller with linguistically defined IF-THEN rules can achieve stable level regulation with smooth transient behavior, outperforming simple on/off control and providing a viable, model-free alternative to classical PID controllers for this non-linear system (Ross, 2010; Passino & Yurkovich, 1998).

4. Labview VI

4.1 Fuzzy System Designer – Variables and Membership Functions

The fuzzy system was designed using LabVIEW's Fuzzy System Designer (tank_fuzzy.fs). Two input variables were defined: dH/dT (the rate of change of tank height) and Tank Error (the deviation from the desired setpoint). Each input variable is described by five triangular membership functions covering the linguistic terms: Fast Negative, Slow Negative, Zero, Slow Positive, and Fast Positive. The single output variable, Flow Change, is described by seven triangular membership functions: Large Negative (Lg neg), Medium Negative (Med neg), Small Negative (Sm neg), No Change, Small Positive (Sm pos), Medium Positive (Med pos), and Large Positive (Lg pos).

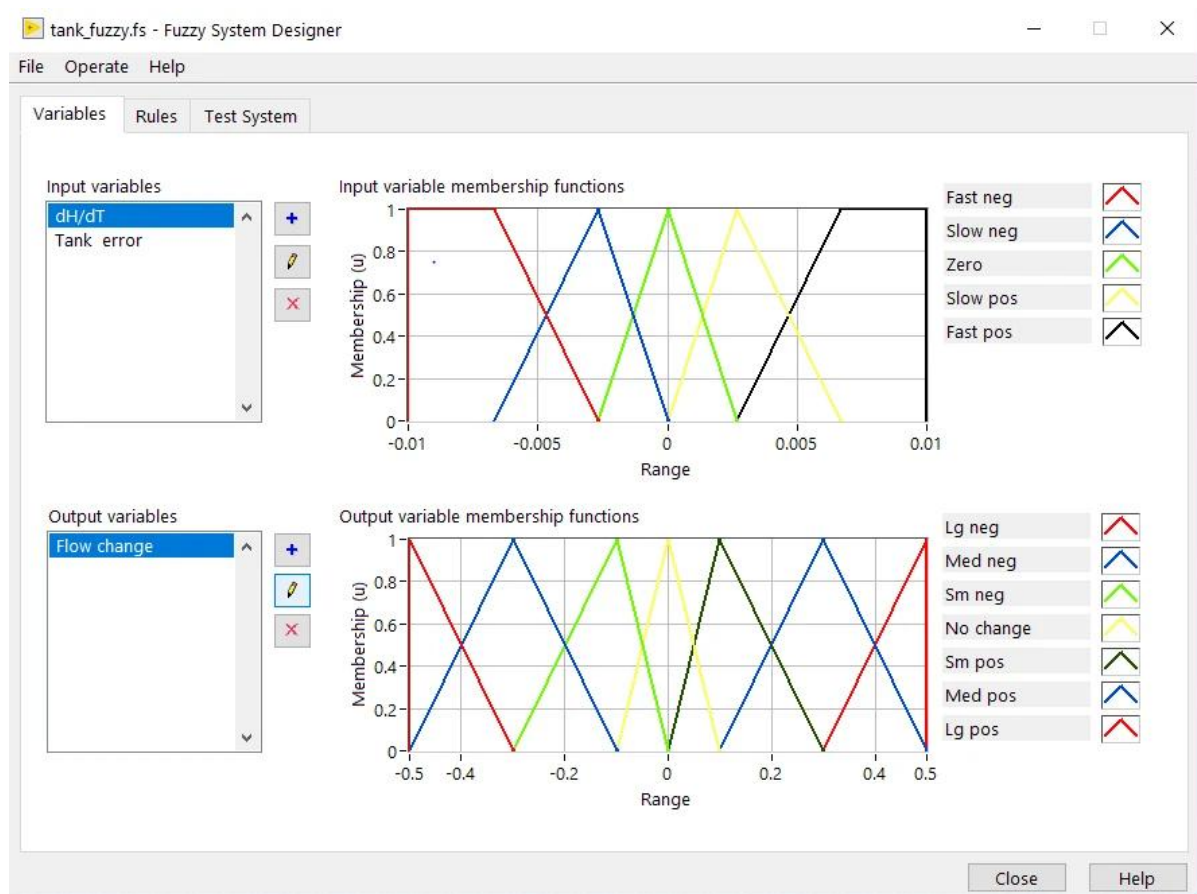


Figure 2: Fuzzy System Designer – Input and Output Variable Membership Functions

4.2 Fuzzy Rule Base

The rule base was constructed using AND (Minimum) antecedent connectives and Minimum consequent implication. The rules map combinations of dH/dT and Tank Error linguistic states to appropriate Flow Change outputs. The complete set of fuzzy rules governing the system is as follows:

1. IF ' dH/dT ' IS 'Fast neg' AND 'Tank error' IS 'Lg neg' THEN 'Flow change' IS 'No change'
2. IF ' dH/dT ' IS 'Fast neg' AND 'Tank error' IS 'Sm neg' THEN 'Flow change' IS 'Sm pos'
3. IF ' dH/dT ' IS 'Fast neg' AND 'Tank error' IS 'None' THEN 'Flow change' IS 'Med pos'
4. IF ' dH/dT ' IS 'Fast neg' AND 'Tank error' IS 'Sm pos' THEN 'Flow change' IS 'Lg pos'
5. IF ' dH/dT ' IS 'Fast neg' AND 'Tank error' IS 'Lg pos' THEN 'Flow change' IS 'Lg pos'
6. IF ' dH/dT ' IS 'Slow neg' AND 'Tank error' IS 'Lg neg' THEN 'Flow change' IS 'Sm neg'
7. IF ' dH/dT ' IS 'Slow neg' AND 'Tank error' IS 'Sm neg' THEN 'Flow change' IS 'No change'
8. IF ' dH/dT ' IS 'Slow neg' AND 'Tank error' IS 'None' THEN 'Flow change' IS 'Sm pos'
9. IF ' dH/dT ' IS 'Slow neg' AND 'Tank error' IS 'Sm pos' THEN 'Flow change' IS 'Med pos'
10. IF ' dH/dT ' IS 'Slow neg' AND 'Tank error' IS 'Lg pos' THEN 'Flow change' IS 'Lg pos'
11. IF ' dH/dT ' IS 'Zero' AND 'Tank error' IS 'Lg neg' THEN 'Flow change' IS 'Lg neg'
12. IF ' dH/dT ' IS 'Zero' AND 'Tank error' IS 'Sm neg' THEN 'Flow change' IS 'Sm neg'
13. IF ' dH/dT ' IS 'Zero' AND 'Tank error' IS 'None' THEN 'Flow change' IS 'No change'
14. IF ' dH/dT ' IS 'Zero' AND 'Tank error' IS 'Sm pos' THEN 'Flow change' IS 'Sm pos'
15. IF ' dH/dT ' IS 'Zero' AND 'Tank error' IS 'Lg pos' THEN 'Flow change' IS 'Lg pos'
16. IF ' dH/dT ' IS 'Slow pos' AND 'Tank error' IS 'Lg neg' THEN 'Flow change' IS 'Lg neg'
17. IF ' dH/dT ' IS 'Slow pos' AND 'Tank error' IS 'Sm neg' THEN 'Flow change' IS 'Med neg'
18. IF ' dH/dT ' IS 'Slow pos' AND 'Tank error' IS 'None' THEN 'Flow change' IS 'Sm neg'
19. IF ' dH/dT ' IS 'Slow pos' AND 'Tank error' IS 'Sm pos' THEN 'Flow change' IS 'No change'
20. IF ' dH/dT ' IS 'Slow pos' AND 'Tank error' IS 'Lg pos' THEN 'Flow change' IS 'Sm pos'
21. IF ' dH/dT ' IS 'Fast pos' AND 'Tank error' IS 'Lg neg' THEN 'Flow change' IS 'Lg neg'
22. IF ' dH/dT ' IS 'Fast pos' AND 'Tank error' IS 'Sm neg' THEN 'Flow change' IS 'Lg neg'
23. IF ' dH/dT ' IS 'Fast pos' AND 'Tank error' IS 'None' THEN 'Flow change' IS 'Med neg'
24. IF ' dH/dT ' IS 'Fast pos' AND 'Tank error' IS 'Sm pos' THEN 'Flow change' IS 'Sm neg'
25. IF ' dH/dT ' IS 'Fast pos' AND 'Tank error' IS 'Lg pos' THEN 'Flow change' IS 'No change'

Defuzzification is performed using the Center of Area method, which computes the centroid of the aggregate output membership function to yield a crisp flow change value.

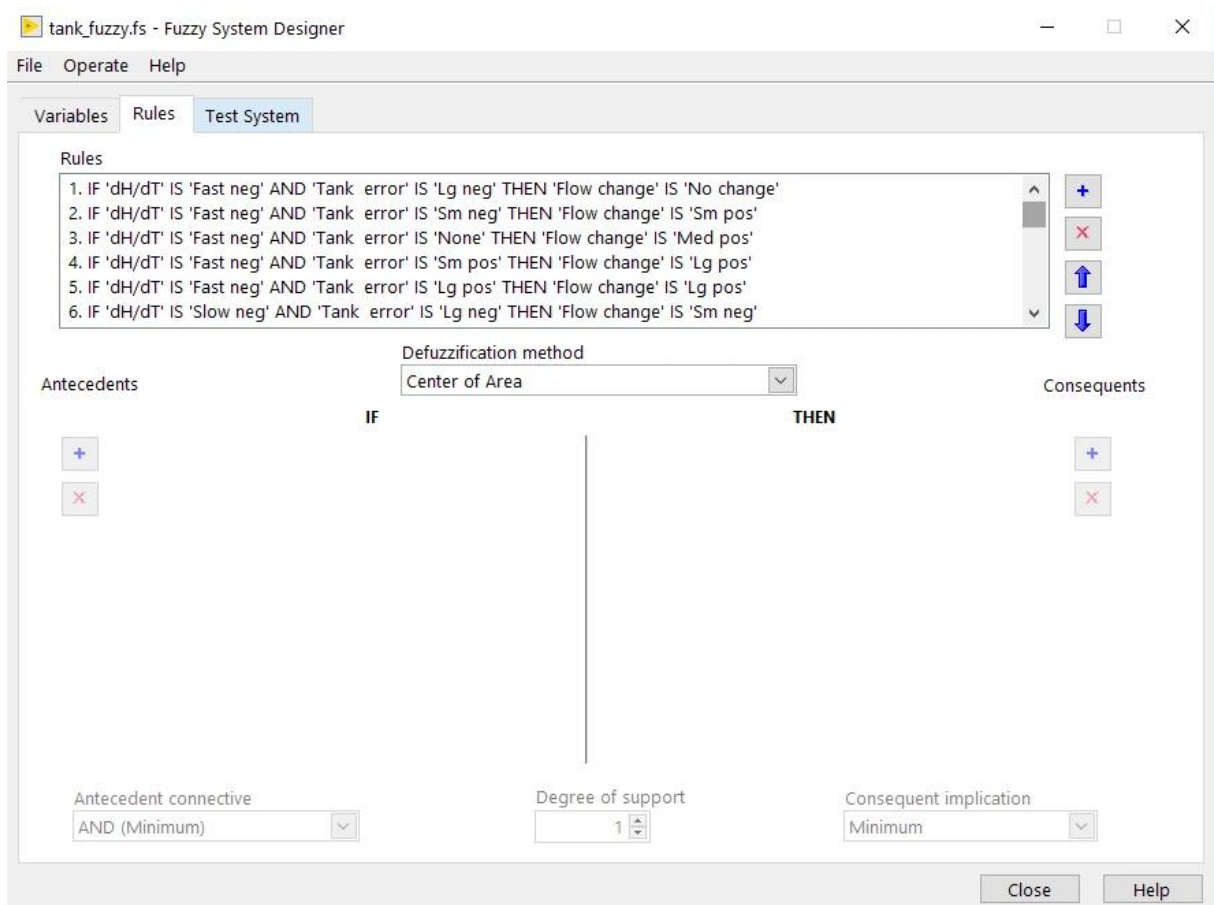


Figure 3: Fuzzy System Designer – Rule Base Definition

4.3 Front Panel Interface & Real-Time Monitoring

The Front Panel serves as the human-machine interface (HMI) and real-time visualization dashboard. It features a dynamic tank graphic, setpoint controls, a flow rate gauge, and a Process Monitor graph that plots the actual water level, flow rate, and target setpoint simultaneously. Additionally, a rule weights array at the bottom displays the activation degree of each fuzzy rule in real time, providing transparency into the controller's decision logic during transient and steady-state phases.

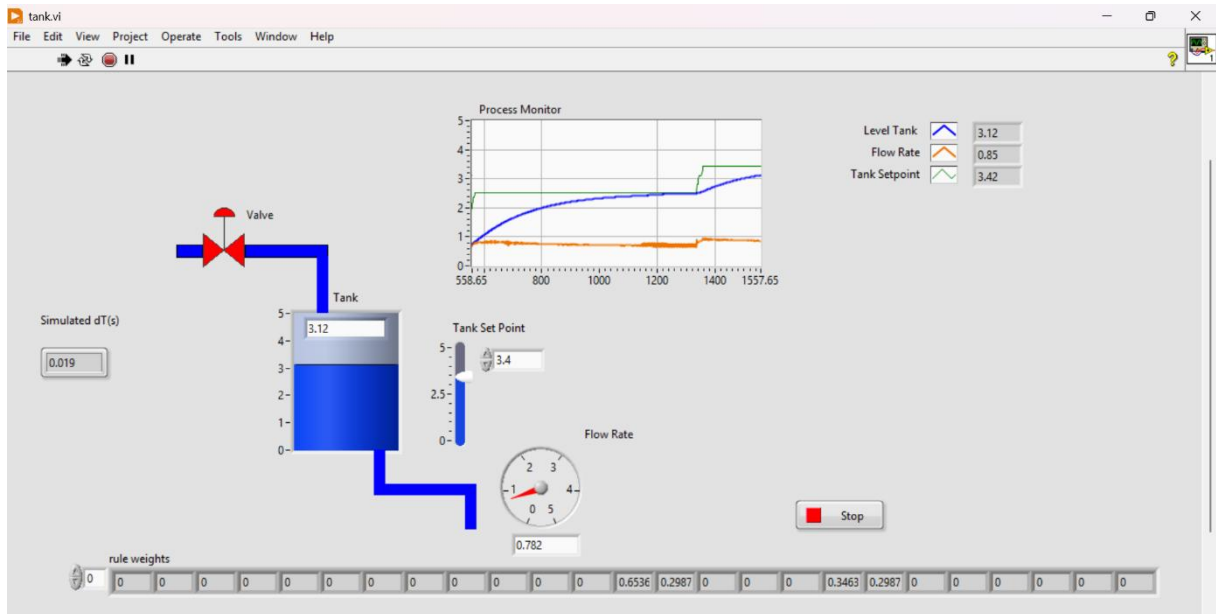


Figure 4: LabVIEW Front Panel featuring tank visualization

4.4 Block Diagram Architecture & Implementation

The Block Diagram implements a deterministic real-time control loop executing at a simulated time step of $dT \approx 0.019$ s. The FL Fuzzy Controller VI reads the fuzzified inputs, evaluates the rule base, and outputs the required flow change. This value drives a MathScript Node that solves the tank's governing differential equations (Torricelli's law for outflow and mass balance for height update). A conditional floor clamp prevents negative water levels. The computed height is fed back to the controller input, closing the feedback loop. Timing is managed via a simulated clock and wait function to ensure synchronized, real-time execution.

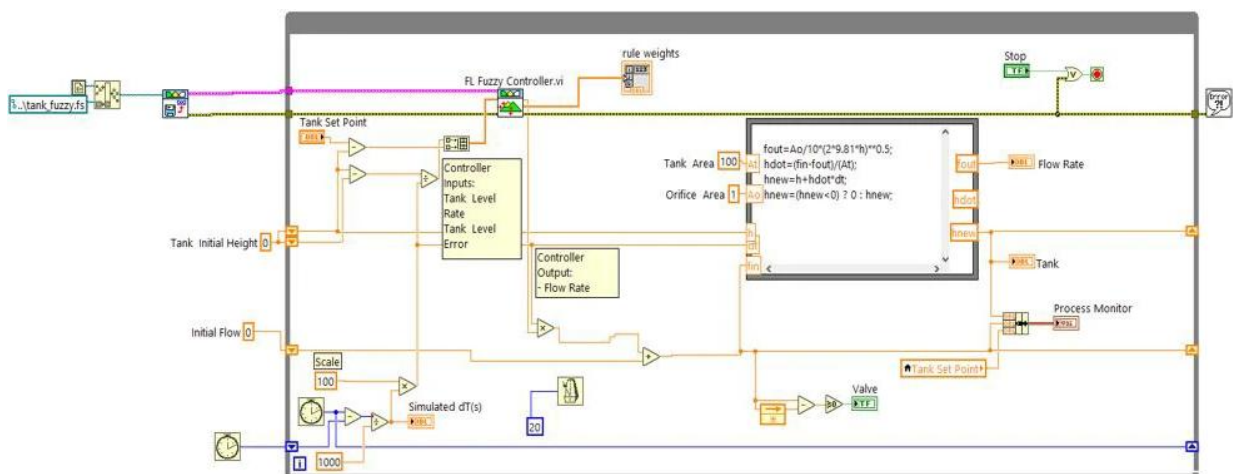


Figure 5: LabVIEW Block Diagram – Fuzzy Logic Tank Control System

5. Results

The simulation was executed with a target tank setpoint of 2.50 units, starting from an initial water level of 0 units. The system parameters included a tank area (A_t) of 100, an orifice area (A_o) of 9, and a simulation time step (dT) of approximately 0.019 seconds.

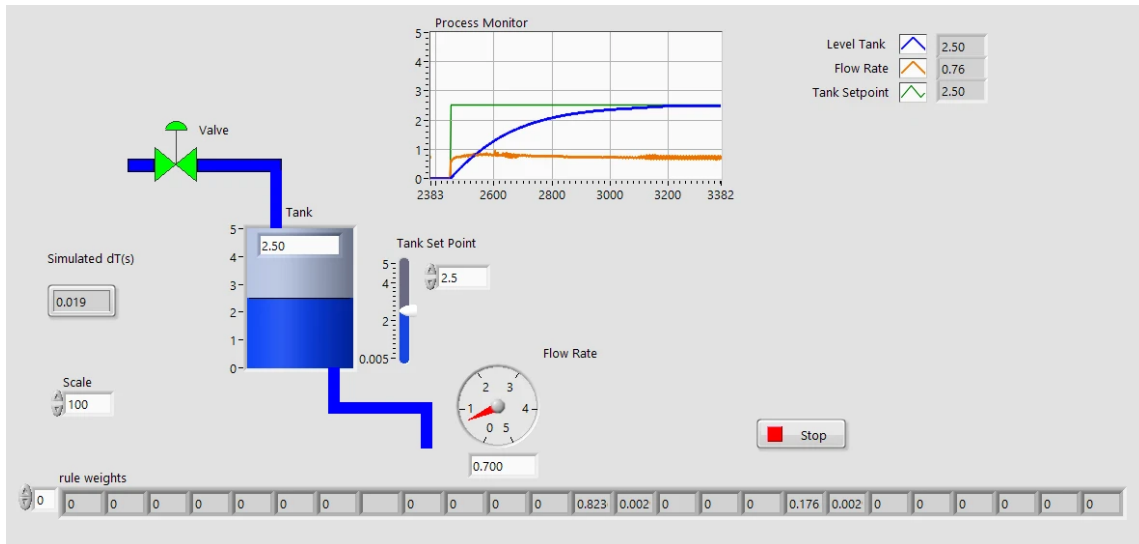


Figure 6: LabVIEW Front Panel – Tank Level Regulated

The results recorded by the Process Monitor graph (Figure 6) demonstrate a highly effective control response. As observed, the tank level (blue trace) rises dynamically from the initial state and converges smoothly to the 2.50 setpoint. Notably, the response exhibits no overshoot or oscillations, indicating that the fuzzy controller effectively dampened the non-linear dynamics of the filling process. Simultaneously, the flow rate (orange trace) initially spikes to rapidly fill the tank and then gradually decreases, stabilizing at approximately 0.76 units. This steady-state flow precisely compensates for the gravitational outflow through the orifice, satisfying the mass balance requirements.

A significant discovery during the analysis was the real-time visualization of the fuzzy inference process via the Rule Weights array located at the bottom of the Front Panel. This array provided transparency into the controller's decision-making logic. During the transient filling phase, weights were distributed among rules corresponding to "negative" errors and "fast" rates. However, as the tank level stabilized at the 2.50 setpoint, the weight for the rule associated with zero error (specifically the 13th rule: "If dH/dT is Zero and Tank Error is None...") surged to a dominant value of approximately 0.823, while other weights diminished to near zero (e.g., 0.002). This confirms that the system correctly identified the equilibrium state and ceased aggressive correction, transitioning to a maintenance mode.

Significance of Findings:

The results validate the hypothesis that a fuzzy logic controller can manage non-linear fluid systems effectively without a precise mathematical model. The smooth convergence and accurate steady-state regulation demonstrate the superiority of the linguistic IF-THEN rule base in handling uncertainty compared to rigid control methods. Furthermore, the real-time rule weight data proves that the Center of Area (Centroid) defuzzification method successfully produced a crisp, stable output that eliminated chattering, ensuring the physical valve would remain in a steady position during steady-state operation.

6. Discussion / Conclusion

The simulation results conclusively validate the central hypothesis that a linguistically defined fuzzy logic controller can effectively regulate water tank levels without requiring precise mathematical modeling. By achieving stable setpoint tracking with zero overshoot and negligible steady-state error, the system directly answers the primary research question: fuzzy inference successfully substitutes rigid differential equations with intuitive IF-THEN rules, handling non-linear fluid dynamics through expert-derived linguistic mappings rather than analytical derivation.

The significance of these findings extends to multiple stakeholders. For industrial process engineers and water management operators, the results demonstrate that fuzzy control offers a robust, model-free alternative to conventional PID tuning, particularly in systems with uncertain parameters or time-varying dynamics. For control systems researchers, the transparent rule-weight visualization confirms that fuzzy architectures provide interpretable decision-making pathways, bridging the gap between black-box AI controllers and traditional mathematical models. Furthermore, the smooth convergence behavior highlights the practical advantage of overlapping membership functions in eliminating control chattering, a critical requirement for prolonging the lifespan of physical actuators such as industrial valves.

From an implementation perspective, the evaluation of LabVIEW's Fuzzy Logic Toolkit confirms its viability as a rapid prototyping environment for real-time intelligent control. The graphical Fuzzy System Designer significantly reduced development time by allowing intuitive configuration of membership functions and rule bases without manual coding. The successful integration of the .fs configuration file with the FL Fuzzy Controller VI and the MathScript-based tank dynamics node validated the toolkit's modular architecture. Real-time execution at a 19 ms sampling interval further proved that the computational overhead of fuzzy inference remains well within standard industrial control constraints.

In conclusion, this project successfully demonstrates that fuzzy logic provides a highly effective, transparent, and adaptable framework for non-linear process control. The designed controller not only met all performance criteria but also showcased the practical synergy between intelligent control theory and LabVIEW's real-time simulation capabilities. These findings reinforce the viability of fuzzy systems as a scalable solution for modern automation challenges, offering a pathway toward more resilient, operator-friendly control strategies in water distribution and similar industrial applications.

7. References

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